UTPAL SONOWAL

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SUMMARY

Full-stack developer focused on building secure, scalable web applications and AI-powered solutions using TypeScript, Node.js, PostgreSQL, and Redis. Experienced in REST APIs, authentication, role-based access control, and automating business workflows.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C/C++
Frontend: React, Next.js, Tailwind CSS, Bootstrap, Shadcn UI, React Router, Redux Toolkit, Context API, Axios
Backend: Node.js, Express.js, FastAPI, JWT Authentication, Authentication & Authorization, Zod
Databases: SQL**,** PostgreSQL, MongoDB, Prisma ORM, Redis
AI/ML: LLMs, RAG, Prompt Engineering, Gemini API, Embeddings, Vector Databases
DevOps: AWS (EC2, S3, Lambda), Docker, CI/CD, Github Actions, Vercel, Render
Tools: Git, GitHub, Postman, Figma, VS Code

EXPERIENCE

Full Stack Developer Intern *Jan 2025 – Dec 2025*
National Informatics Centre (NIC) — India

- Developed a two-tier transportation management system to automate transportation billing across warehouses, wholesalers, retailers, and commodity distribution.
- Automated transportation cost and billing calculations based on commodity, distance, and geographical area, eliminating complex manual calculations**.**
- Reduced billing preparation time from approximately 3–4 hours to around 10 minutes by automating data processing and bill generation.
- Generated downloadable Excel billing sheets with automated formatting based on predefined templates and designs, eliminating repetitive manual spreadsheet preparation.

PROJECTS

LnkShrt — Next Gen URL Shortener & Analytics Live Demo | GitHub
Node.js, Express, PostgreSQL, Redis, TypeScript, Next.js

- Built a full-stack URL management platform supporting custom aliases or auto-generated short links, with password protection, UTM parameters, link expiration, QR codes, and link management.
- Engineered a free analytics system that tracks link performance across clicks, location, device, browser, operating system, and referrer, providing detailed traffic insights.
- Implemented secure authentication with JWT access/refresh tokens and Redis + BullMQ for queued email processing, enabling automated email delivery without blocking user requests.
- Load-tested the URL redirect API using k6 (100 VUs, 18,000 requests): 20.68 ms p95 latency at 200 RPS, 0% HTTP failures.

Study Buddy — Full-Stack RAG Chatbot with Production-Grade Authentication Live Demo | Github
FastAPI, PostgreSQL, pgvector, Redis, Gemini, TypeScript, React.js

- Designed and built a production-grade auth system with email/OTP + Google/GitHub OAuth, including account linking, email verification.
- Implemented secure session management using access/refresh token rotation with reuse detection, HttpOnly/Secure cookies, and full session revocation
- Developed a RAG pipeline (PostgreSQL + pgvector) enabling users to query their personal notes with cited, source-linked LLM-generated answers.
- Enforced multi-tenant data isolation so retrieval and queries never cross user boundaries
- Built background job infrastructure (Redis) for async embedding generation and retry-safe transactional email delivery

EDUCATION

Bachelor of Computer Applications (BCA) 2023 – 2026
North Lakhimpur University
CGPA: 7.8 / 10